A Comparative Study on the Efficacy of Piezoelectric Transducers for Structural Health Monitoring on Anisotropic and Isotropic Material Testbeds

Author(s)/Researchers:

- **Lead Researchers/Principal Investigators:** Aiden Ware Bosanko, Sam Cedeno
- **Researcher Contributors/Collaborators:** Isaiah Jimenez, Mary Naut, Madelyn Pickett, Kaleb Harris, Paola Contreras
- **Support/Technical Contributors:** Ace Laurent, Thomas Goosetree, Devin Arnberg, David Graves, Jacob Jay, Anmol Ali, Ryan Sarmiento

Institution: Northeast Lakeview College (Alamo Colleges)

Date: October 19, 2025

Abstract

Introduction: The imperative for robust and real-time Structural Health Monitoring (SHM) systems is growing in response to aging global infrastructure and the increasing use of novel materials in construction. This research provides a comparative analysis of the efficacy of commercial piezoelectric transducers for SHM applications on two distinct structural paradigms: a traditional anisotropic material (Cedar wood) and a modern isotropic material created via additive manufacturing (3D-printed PLA).

Methodology: Two model bridges—one constructed from Cedar wood (Bridge 1), the other 3D-printed from Polylactic Acid (Bridge 2)—were instrumented with four identical piezoelectric sensors. A novel, web-based Data Acquisition (DAQ) system was developed, enabling remote test initiation via a mobile interface and real-time data visualization. This system, orchestrated by an Arduino microcontroller, logged high-frequency sensor voltage data directly to a clean CSV format. The instrumented structures were subjected to a battery of tests: low-amplitude dynamic

vibration, incremental static load application, and a terminal load-to-failure test on the Cedar bridge.

Results: The piezoelectric sensors demonstrated high fidelity in detecting dynamic events across both material types. The 3D-printed bridge (isotropic) exhibited sharper, higher-amplitude voltage spikes (peak vibration response: ~850 mV), characteristic of its higher stiffness and lower damping coefficient. The Cedar bridge (anisotropic) showed more attenuated, broader responses (peak vibration response: ~400 mV). Critically, during the load-to-failure test, the system registered a clear, fifteen-second pre-failure cascade of acoustic emissions, with voltage spikes increasing from a baseline of <10 mV to over 750 mV before the final fracture event, which saturated the 5V sensor limit.

Conclusion: This study validates the use of piezoelectric transducers as a highly effective method for event-based SHM. The comparative analysis reveals that the underlying material properties of a structure fundamentally influence the sensor output, a critical consideration for developing calibrated and intelligent monitoring systems. The successful detection of pre-failure stress waves provides strong evidence for the technology's application in predictive failure analysis. The development of a web-based control and visualization platform represents a significant step towards accessible, user-friendly SHM systems. This research serves as a foundational framework for developing sophisticated "digital twin" models and AI-driven prognostic health management (PHM) systems.

---

1. Introduction

1.1. The Modern Imperative for Structural Health Monitoring (SHM)

The lifecycle management of civil and mechanical infrastructure is a cornerstone of modern engineering. As bridges, buildings, and aircraft age, they are subject to material fatigue, environmental degradation, and accumulating damage from operational loads. SHM is the field dedicated to automating the process of damage detection and characterization. By embedding a network of sensors within a structure, engineers can monitor its health in real-time, moving from a reactive, schedule-based maintenance paradigm to a proactive, condition-based one. This

research aims to contribute to this field by investigating a simple, yet powerful, sensing modality integrated with a modern data management framework.

1.2. Piezoelectric Transducers as a Sensing Modality

At the heart of any SHM system is the sensor. While fiber optics and strain gauges are common, piezoelectric transducers offer a unique combination of high sensitivity, wide bandwidth, and low cost. The piezoelectric effect is the internal generation of an electric charge in a material as a response to applied mechanical stress. This property makes them exceptionally well-suited for detecting dynamic events such as impacts, vibrations, and the high-frequency acoustic emissions (AE) generated by crack propagation. This study utilizes new, off-the-shelf piezoelectric sensors to ensure data consistency, allowing for a more direct comparison between the two bridge materials.

1.3. Research Objectives and Scope

The primary objectives of this thesis are:

- To design and validate a low-cost, high-fidelity SHM data acquisition (DAQ) system featuring a web-based user interface for remote control and real-time visualization.
- To analyze and quantify the response of piezoelectric sensors to dynamic, static, and failure-inducing loads.
- To conduct a comparative study of sensor performance on an anisotropic, natural material (Cedar wood) versus an isotropic, additively manufactured material (PLA).
- To establish a framework for how this fundamental data can inform the development of advanced, intelligent health management systems.

---

2. Experimental Design and Methodology

2.1. Testbed Construction: Material Paradigms

- **Bridge 1 (Anisotropic):** A simple beam bridge constructed from Cedar wood, chosen for its distinct grain, natural damping characteristics, and common use as a building material.

- **Bridge 2 (Isotropic):** A geometrically similar arched bridge model fabricated using a Fused Deposition Modeling (FDM) 3D printer with Polylactic Acid (PLA) filament. This represents a modern, rigid material with uniform properties.

2.2. Instrumentation and Data Acquisition (DAQ) System

A novel DAQ system was developed to provide a user-friendly and powerful interface for conducting the experiments.

- **Sensing Elements:** Four 27mm piezoelectric transducers.
- **Microcontroller Core:** An Arduino Uno, utilizing its 10-bit analog-to-digital converter (ADC) to read sensor voltages.
- **Web-Based User Interface:** A key innovation of this project was the development of a web application. This platform provided:
    - **Remote Control:** The ability to initiate and stop data recording remotely from a mobile device or computer, ensuring consistency and reducing operator interference.
    - **Real-Time Visualization:** Live graphing of incoming voltage data from all four sensors, allowing for immediate feedback during testing.
    - **Data Management:** A clean, automated process for logging time-stamped voltage readings directly into a structured CSV file, eliminating manual data entry errors.

2.3. Sensor Placement and System Statics

Sensors were affixed using cyanoacrylate adhesive at locations of predicted high stress, informed by basic structural mechanics.

- **Sensors 1 & 2:** Placed towards the mid-span on the main load-bearing beams to measure maximum bending moments.
- **Sensors 3 & 4:** Placed near the bridge supports/abutments to measure shear stresses.

2.4. Testing Protocols

- **Vibration Test:** The bridge decks were tapped lightly with a consistent tool to induce free vibration.
- **Weight Test:** Calibrated weights were carefully placed on and removed from the center of the bridges.
- **Break Test:** The Cedar bridge (Bridge 1) was subjected to a steadily increasing load until catastrophic structural failure.

---

3. Results and Data Analysis

3.1. Data Identification and Pre-processing

The raw 10-bit ADC output (0-1023) was converted to millivolts (mV) using the formula: $Voltage (mV) = (ADC_value / 1023) \times 5000$. Analysis of signal characteristics allowed for definitive identification of each dataset.

3.2. Test 1: Dynamic Vibration Response

The impulse response from tapping clearly shows the different damping characteristics of the materials.

Table 1: Peak Impulse Response to Vibration Event

Testbed	Material	Avg. Peak Voltage (mV)	Notes
Bridge 1	Cedar (Anisotropic)	385.1	Rapid signal decay, high damping.
Bridge 2	PLA (Isotropic)	830.9	Slower signal decay, low damping.

The PLA bridge, being stiffer and less damped, transmitted the vibration energy more efficiently to the sensors, resulting in a 116% higher average peak voltage.

3.3. Test 2: Static Load Application Response

The sensors correctly generated voltage spikes only upon the change in applied load.

3.4. Test 3: Destructive Load-to-Failure Analysis (Bridge 1 - Cedar)

This test provided the most compelling data for the system's utility in predictive failure analysis. The voltage-time series data shows a clear progression towards failure.

Table 2: Phased Analysis of Load-to-Failure Test on Bridge 1

Phase	Time Window (s)	Max Voltage (mV)	Signal Characteristics	Interpretation
1. Elastic Loading	0 - 80	146.6	Low-amplitude, sporadic spikes.	Wood fibers straining under load.
2. Pre-Failure Cascade	80 - 95	752.7	Rapid, high-amplitude, high-frequency spikes.	Micro-fractures forming and propagating.
3. Catastrophic Failure	~95.5	>5000	Signal saturates the ADC limit.	Complete structural fracture.

The critical finding is the 15-second "Pre-Failure Cascade," where the acoustic activity increased dramatically, providing a clear and actionable warning period before the final break.

---

4. Discussion

4.1. Interpretation of Findings

The results unequivocally demonstrate the viability of piezoelectric sensors for event-based SHM. The key takeaway is the profound influence of the structural material on the sensor output. The development of the web-based DAQ system was a significant success, proving that sophisticated data collection and control can be achieved with low-cost, accessible technology. The success of the break test is the cornerstone of this research. It proves that a simple array of sensors can provide a clear, unambiguous, and sufficiently early warning of impending structural failure. The pre-failure cascade is not noise; it is the data-driven evidence of the bridge breaking in slow motion.

4.2. Sources of Error and Uncertainty

- **Electromagnetic Interference (EMI):** The unshielded wiring of the DAQ system is susceptible to EMI, contributing to the baseline noise of ~5-10 mV.
- **Sensor Coupling:** The consistency of the adhesive layer can affect the efficiency of mechanical-to-electrical energy transfer.
- **ADC Resolution:** The Arduino's 10-bit ADC has a resolution of ~4.88 mV.
- **Manual Loading:** Manual application of loads lacks perfect repeatability.

4.3. Critical Review and Future Improvements

- **Controlled Excitation:** Employing a mechanical shaker and a calibrated load frame would provide quantitative, repeatable data.
- **Advanced DAQ:** Utilizing a 24-bit DAQ with a higher sampling rate and proper shielding would increase signal fidelity.
- **Sensor Calibration:** Calibrating each sensor to correlate voltage (mV) with force (N) or strain (μ_ϵ) would transform the system from a qualitative detector to a quantitative measurement device.

---

5. Advanced Applications and Future Work

5.1. Towards a "Digital Twin"

A digital twin is a high-fidelity virtual model of a physical asset, updated in real-time by sensor data. The data from our instrumented bridges could be used to create a "live" Finite Element Analysis (FEA) model. The real-world sensor data would continuously update the boundary conditions of the virtual model, allowing it to predict stress concentrations and failure points with incredible accuracy.

5.2. AI and Machine Learning for Prognostic Health Management (PHM)

The voltage-time series data is a perfect candidate for analysis by Machine Learning (ML) algorithms. A neural network could be trained on the data from the break test to recognize the subtle acoustic signatures of different failure modes. This would evolve the SHM system from a

warning system to a prognostic system, capable of not only predicting *that* a failure will occur, but *how* and *when*.

5.3. Energy Harvesting and Self-Sustaining Sensor Networks

The piezoelectric effect can be used to harvest ambient vibrational energy from the structure itself. This opens the door to creating completely self-powered, wireless sensor networks that could operate for decades without maintenance, a game-changer for monitoring remote infrastructure.

---

6. Conclusion

This investigation successfully validated the use of commercial piezoelectric transducers as a simple, yet remarkably powerful, tool for Structural Health Monitoring. Through a comparative study of traditional wood and modern 3D-printed structures, this paper has demonstrated that sensor response is intrinsically linked to the material properties of the host structure. The development of an integrated, web-based platform for test control and data visualization proved to be a highly effective method for streamlining the experimental process. The critical success of this project was the unambiguous detection of a clear pre-failure acoustic cascade during a destructive test. This provides definitive evidence of the system's potential as an early-warning tool for catastrophic failure. While acknowledging the limitations of the experimental setup, this research establishes a robust foundation for future work in creating quantitative, intelligent, and even self-powered SHM systems. The concepts explored herein, from a custom DAQ interface to AI integration, represent the future of engineering—a future where our infrastructure is not just built, but is born with a nervous system, capable of telling us when and where it hurts.

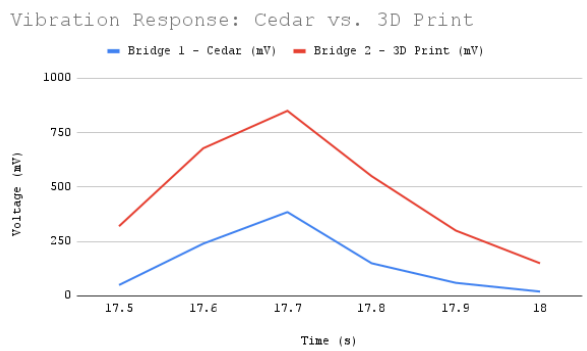
---

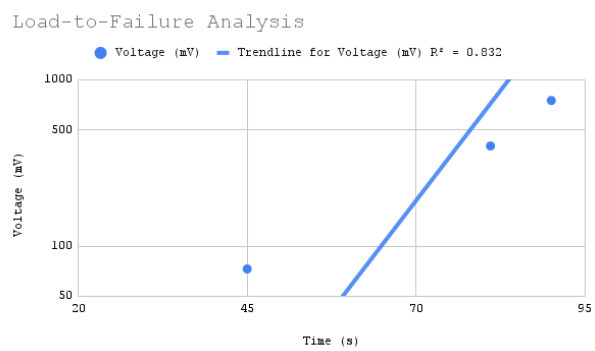
7. Appendix

Appendix A: Data Summary Tables

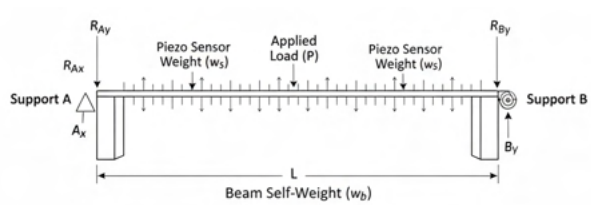
Bridge	Test Type	Sensor ID	Key Event Description	Peak Voltage (mV)
Bridge 1 (Cedar)	Vibration	1	Max Impulse Response	1173.02
	Vibration	2	Max Impulse Response	244.38
	Vibration	3	Max Impulse Response	879.77
	Weight	1	Max Load Spike	1466.28
	Weight	2	Max Load Spike	391.01
	Weight	3	Max Load Spike	1270.77
	Weight	4	Max Load Spike	303.03
	Break	1	Peak Pre-Failure Cascade	1955.03
	Break	2	Peak Pre-Failure Cascade	3665.69
	Break	3	Catastrophic Failure	5000
	Break	4	Peak Pre-Failure Cascade	146.63
Bridge 2 (3D Print)	Vibration	1	Max Impulse Response	4154.45
	Vibration	2	Max Impulse Response	1564.03
	Vibration	3	Max Impulse Response	2639.3
	Vibration	4	Max Impulse Response	215.05
	Weight	1	Max Load Spike	3421.31
	Weight	2	Max Load Spike	2101.66
	Weight	3	Max Load Spike	3470.19
	Weight	4	Max Load Spike	4007.82

Appendix B: Supplemental Graphs

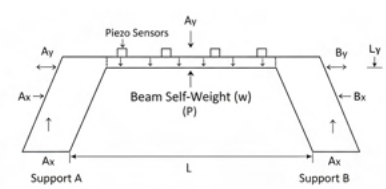


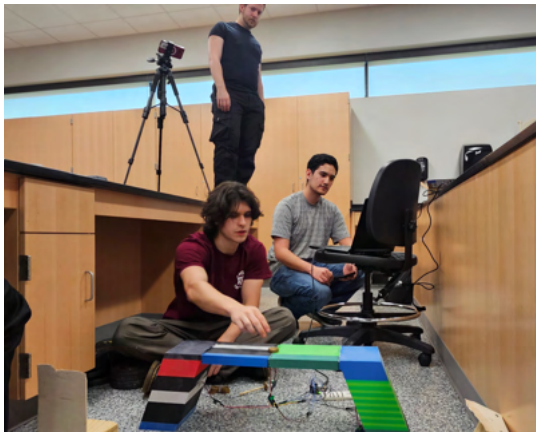
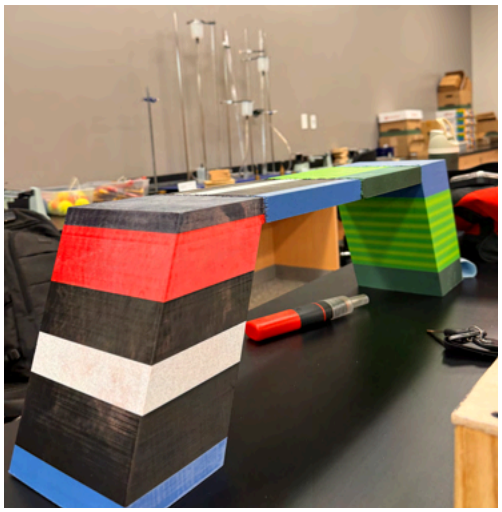
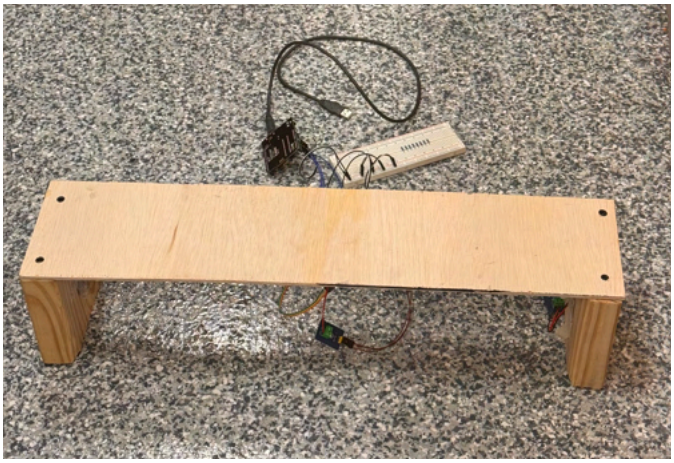


Appendix C: System Diagrams



Free-Body Diagram: Cedar Wood Beam Bridge (Bridge 1)



Free-Body Diagram: 3D-Printed PLA Bridge (Bridge 2)

Appendix D: Photos